

Christian Groeneveld

Radio Astronomer

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Experience

- 2025–Present Postdoc, *INAF - IRA*, Bologna, IT
Postdoc in the Ultra-Low Universe (ULU) group (Francesco de Gasperin). Here I operated as an independent scientist, studying diffuse radio emission in galaxy clusters using state-of-the-art calibration techniques, using different radio telescopes. In addition, I was an integral part of the development of calibration tools currently used in the Library for Low Frequencies (used for the LOFAR LBA Sky Survey)
- 2020–2025 PhD Candidate, *Leiden Observatory*, Leiden, NL
PhD student in the LOFAR group (Reinout van Weeren). My main focus was pioneering research in the most challenging radio calibration regimes, in particular LBA long baselines, and the decameter wavelength range.

Education

- 2018–2020 MSc in Astronomy and Data Science, *Leiden University*, Leiden, NL
2015–2018 BSc in Astronomy, BSc in Physics, *Leiden University*, Leiden, NL

Skills

Key Skills	Radio Astronomy	● ● ● ● ●
	Low-Frequency Radio calibration	● ● ● ● ●
	Radio Polarisation	● ● ● ● ●
Languages	Python, SQL, $\LaTeX$, Lua, bash	

Select Publications

- 2022 **C. Groeneveld**, R.J. van Weeren, G.K. Miley et al. *Pushing sub-arcsecond resolution imaging down to 30 MHz with the trans-European International LOFAR Telescope*, A&A, 658, 9. This paper represents high-resolution observations of several bright radio sources at unprecedented low frequencies (down to 30 MHz.). The high spatial resolution, combined with the unprecedented low radio frequencies require novel calibration techniques in order to be corrected for ionospheric effects. This paper is a fundamental paper for the current development of a pipeline for calibrating LBA long baseline data.
- 2024 **C. Groeneveld**, R.J. van Weeren, E. Osinga et al. *Characterization of the decametre sky at subarcminute resolution*, Nature Astronomy, 8, 786. This paper showcases our successful attempt to open the last electromagnetic wavelength range that has remained unexplored: the decameter wavelength range. Technical improvements, but most importantly, improved calibration techniques have allowed us to map the sky in the decameter band with unprecedented resolution and depth.

Invited talks

- 2022 Harvard Center for Astrophysics, Galaxy Cluster Meeting
- 2024 URSI - RASC - *Decameter astronomy with LOFAR: a new window on the Universe*
- 2024 7th LOFAR Data School - *Calibrating systematic effects*
- 2026 SPARCS XIV - *MeerKAT UHF observations of the Perseus Cluster*
- 2026 8th LOFAR Data School - *Calibration and self-calibration*

Awarded Proposals

- 2020 Extragalactic A-Team sources with LOFAR LBA, *LOFAR - Cycle 15*
- 2026 Pilot study of Perseus galaxy cluster with MeerKAT UHF, *MeerKAT - DDT proposal*

Publications with significant contribution

- 2022 **C. Groeneveld**, R.J. van Weeren, G.K. Miley et al. *Pushing sub-arcsecond resolution imaging down to 30 MHz with the trans-European International LOFAR Telescope*, A&A, 658, 9
- 2022 B. Punsly, **C. Groeneveld**, G.J. Hill et al. *The Energetics of the Central Engine in the Powerful Quasar 3C 298*, AJ, 163, 194
- 2024 **C. Groeneveld**, R.J. van Weeren, E. Osinga et al. *Characterization of the decametre sky at subarcminute resolution*, Nature Astronomy, 8, 786
- 2025 **C. Groeneveld**, R.J. van Weeren, A. Botteon et al. *Serendipitous decametre detection of ultra steep spectrum radio emission in Abell 655*, A&A, 693, 99
- 2026 **C. Groeneveld**, R.J. van Weeren, M.-L. Gendron-Marsolais et al. *Investigating the radio emission in the Perseus cluster with LOFAR sub-80 MHz LBA observations*, A&A, 709, 64
- 2026 C. Cordun, **C. Groeneveld** et al. LOFAR Decameter Sky Survey (LoDeSS DR1) in press.

Other publications as co-author

- 2022 J. Harwood, S. Mooney, L.K. Morabito, et al. *The resolved jet of 3C 273 at 150 MHz*, A&A 658, 8
- 2022 L.K. Morabito, N.J. Jackson, S. Mooney et al. *Sub-arcsecond imaging with the International LOFAR Telescope. I. Foundational calibration strategy and pipeline*, A&A, 658, 1
- 2022 A. Botteon, T.W. Shimwell, R. Cassano et al. *The Planck clusters in the LOFAR sky. I. LoTSS-DR2: New detections and sample overview*, A&A, 660, 78
- 2024 R.J. van Weeren, R. Timmerman, V. Vaidya et al. *LOFAR high-band antenna observations of the Perseus cluster: The discovery of a giant radio halo*, A&A, 692, 12
- 2025 L.K. Morabito, N.J. Jackson, J.M.G.H.J. de Jong et al. *A decade of sub-arcsecond imaging with the International LOFAR Telescope*, Ap&SS, 370, 19
- 2026 T.W. Shimwell, M.J. Hardcastle, C. Tasse et al. *The LOFAR Two-metre Sky Survey: VII. Third Data Release*, A&A, 707, 198
- 2026 K.S.L. Srikanth, A. Botteon, R. Cassano, *Combined LOFAR-uGMRT analysis of the diffuse radio emission in the massive clusters Abell 773 and Abell 1351*, A&A, 707, 368
- 2026 M. Cianfaglione, F. de Gasperin, V. Cuciti, *Two-component large-scale radio emission in Abell 2244*, A&A, 713, 91